

2010
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 15, 17, 21, 23 and 25

Total marks – 100

Section I Pages 2–25

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–32
- Allow about 1 hour and 40 minutes for this part

Section II Pages 27–36

25 marks

- Attempt ONE question from Questions 33–37
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

- 1 The International Space Station orbits Earth at an altitude of approximately 330 km. Another satellite, Meteosat, is in geostationary orbit at an altitude of 36 000 km.

Which of the following correctly compares the orbital velocity and orbital period of these satellites?

	<i>International Space Station</i>	<i>Meteosat</i>
(A)	Greater orbital velocity	Shorter orbital period
(B)	Lesser orbital velocity	Shorter orbital period
(C)	Greater orbital velocity	Longer orbital period
(D)	Lesser orbital velocity	Longer orbital period

- 2 Which of the following best describes Galileo's analysis of projectile motion?

- (A) A projectile launched with a great enough velocity would escape Earth's gravity.
- (B) A projectile would travel in a straight line until it ran out of momentum, then it would fall.
- (C) A projectile launched from the equator towards the east with a great enough velocity would orbit Earth.
- (D) A projectile would travel in a parabolic path because it has constant horizontal velocity and constant vertical acceleration.

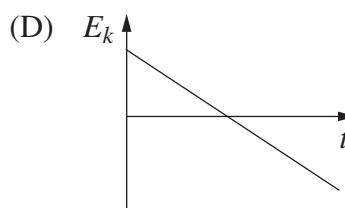
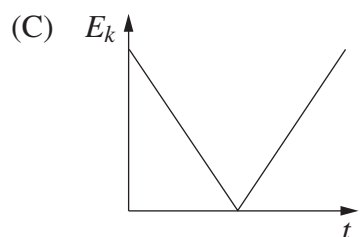
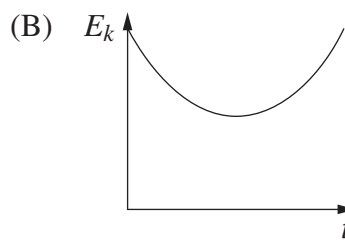
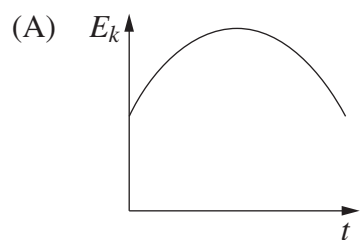
- 3 A scientist at a particle accelerator laboratory observes the lifetime of a particular subatomic particle to be 1.0×10^{-6} s when it is travelling at $0.9999\ c$.

What would the lifetime of the particle be if it were stationary in the laboratory?

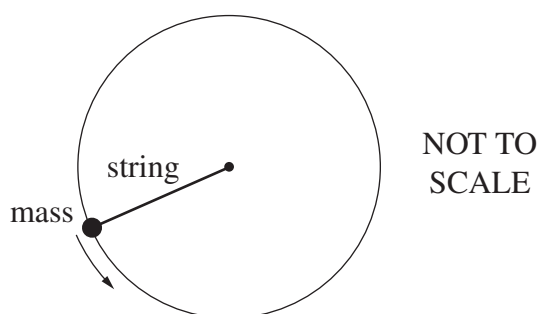
- (A) 1.4×10^{-8} s
- (B) 4.5×10^{-8} s
- (C) 1.0×10^{-6} s
- (D) 7.1×10^{-5} s

- 4 A ball was thrown upward at an angle of 45° . It landed at the same height as thrown.

Which graph best represents the kinetic energy of the ball during its time of flight?



- 5 A 200 g mass is swung in a horizontal circle as shown. It completes 5 revolutions in 3 seconds. The circle has a 2 m diameter.



Which of the following forces is closest to that required to keep the mass moving in this circle?

- (A) 0.50 N
(B) 2.5 N
(C) 10 N
(D) 20 N

- 6 Which statement about the Michelson-Morley experiment is correct?
- (A) It was a valid experiment because it tested the principle of relativity.
 - (B) It was a valid experiment because it took into account the known properties of light.
 - (C) It was an invalid experiment because it did not take into account the particle nature of light.
 - (D) It was an invalid experiment because the speed of Earth through the aether was not taken into account.

- 7 The acceleration due to gravity on the surface of Mercury is 3.6 m s^{-2} .

How much does a 2.0 kg brick weigh on Earth and on Mercury?

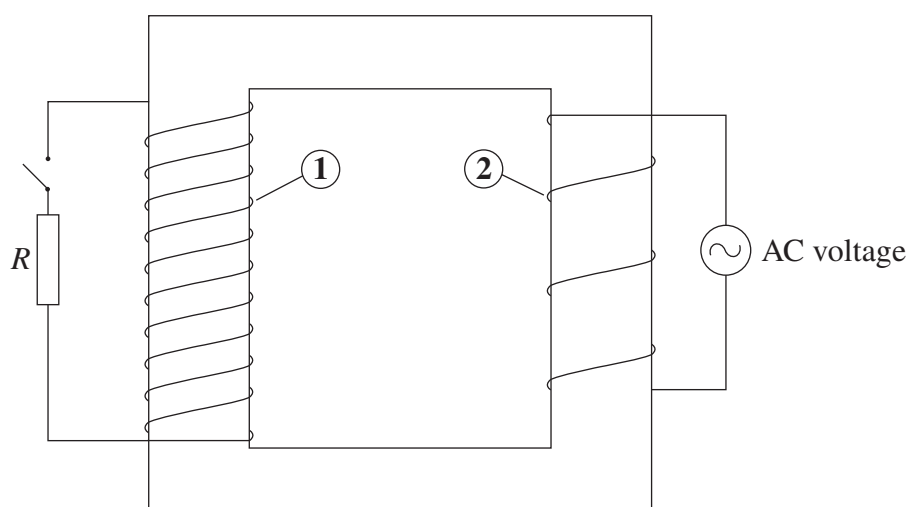
	<i>Weight of brick on Earth</i>	<i>Weight of brick on Mercury</i>
(A)	2.0 kg	2.0 kg
(B)	19.6 kg	7.2 kg
(C)	19.6 N	19.6 N
(D)	19.6 N	7.2 N

- 8 While drilling into a tough material, the DC motor in an electric drill is slowed significantly. This causes its coils to overheat.

Why do the coils overheat?

- (A) The resistance of the coils increases significantly.
 - (B) The increased friction on the drill is converted to heat.
 - (C) The back emf decreases and so the current in the coils increases.
 - (D) The induced eddy currents increase and so more heat is produced.
- 9 Why is high voltage used to transmit electrical energy from power stations to users?
- (A) It helps to protect the system from lightning strikes.
 - (B) It allows the supporting structures to have smaller insulators.
 - (C) It minimises the effects of the electrical resistance of the wires.
 - (D) It ensures that, even with voltage losses, 240 V will still reach the user.

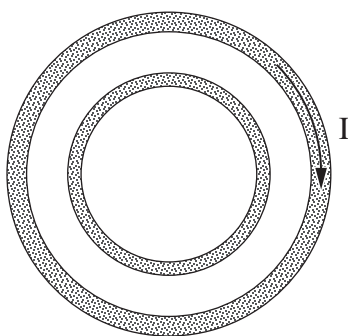
- 10 The diagram shows a model of a transformer in a circuit.



Which of the following correctly identifies Part 1 and Part 2 and the function of this transformer?

	<i>Part 1</i>	<i>Part 2</i>	<i>Function of transformer</i>
(A)	Primary coil	Secondary coil	Step-up
(B)	Secondary coil	Primary coil	Step-down
(C)	Primary coil	Secondary coil	Step-down
(D)	Secondary coil	Primary coil	Step-up

- 11 Two copper rings lie in the same plane as shown.

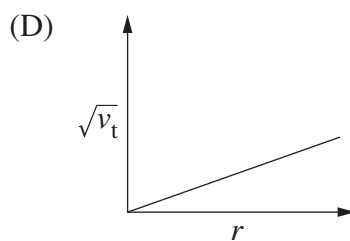
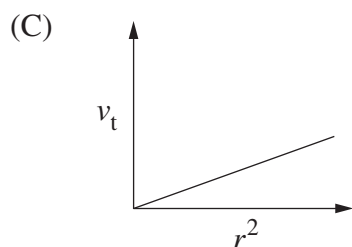
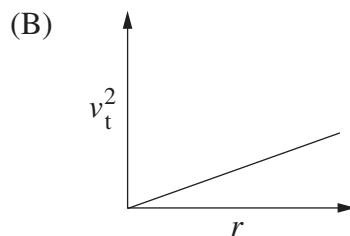
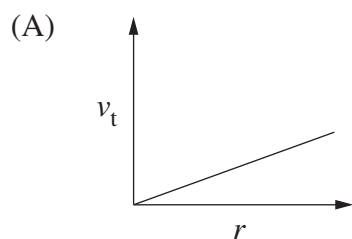


An increasing current flows clockwise around the outer ring.

What happens in the inner ring?

- (A) A decreasing clockwise current flows.
 - (B) A decreasing anticlockwise current flows.
 - (C) An increasing clockwise current flows.
 - (D) An increasing anticlockwise current flows.
- 12 The terminal velocity (v_t) of a spherical object in Earth's atmosphere is proportional to the square root of its radius (r).

Which graph correctly shows this relationship?

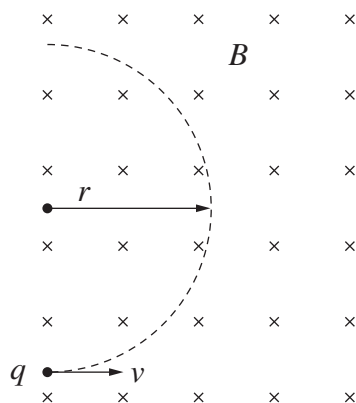


- 13** What was Max Planck's contribution to the development of quantum physics?
- (A) He combined the quantised wave and particle models of light.
 - (B) He analysed the photoelectric effect and described light as quantised energy packets.
 - (C) He explained black body radiation and the photoelectric effect using quantised energy.
 - (D) He hypothesised that the radiation emitted and absorbed by the walls of a black body cavity is quantised.
- 14** Heinrich Hertz devised and performed an experiment to investigate electromagnetic waves. In this experiment he was able to determine the speed of the waves.

Which method was used to determine the speed?

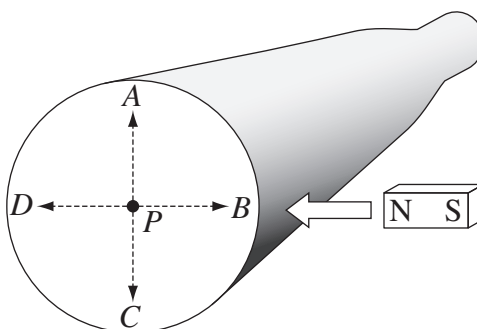
- (A) Timing how long it took the wave to travel a known distance
- (B) Producing a wave of known wavelength and using reflection to determine the frequency
- (C) Producing a wave of known frequency and using interference to determine the wavelength
- (D) Using an interference pattern to determine the distance travelled and time taken

- 15 A charged particle, q , enters a uniform magnetic field B at velocity v . The particle follows a circular path of radius r as shown.



If the magnitude of the magnetic field were doubled and the other variables were kept constant, what would the new radius be?

- (A) $\frac{r}{4}$
- (B) $\frac{r}{2}$
- (C) $2r$
- (D) $4r$
- 16 A cathode ray beam strikes the screen at point P , producing a bright spot. The north end of a magnet is brought towards the beam as shown.



Towards which point does the bright spot move?

- (A) A
- (B) B
- (C) C
- (D) D

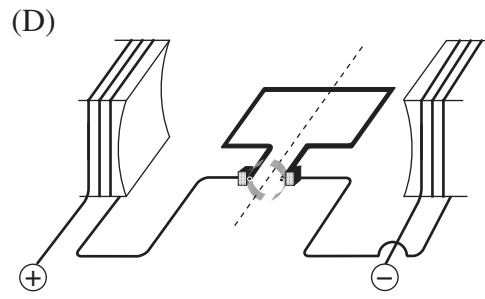
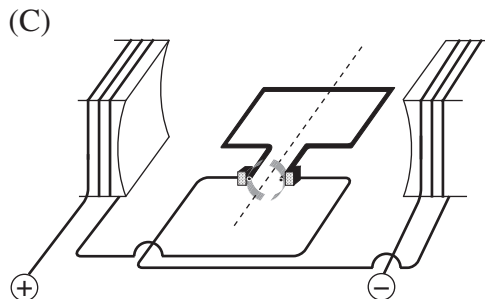
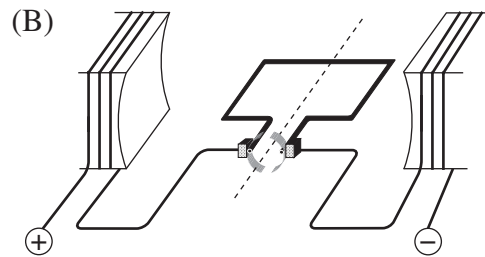
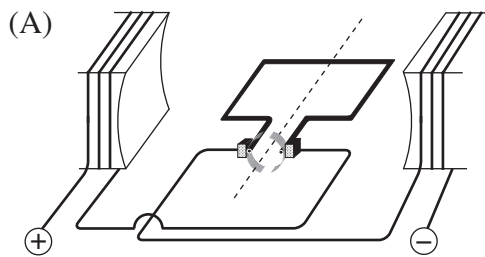
- 17** JJ Thomson determined the charge/mass ratio of the electron by constructing a device which contained
- (A) perpendicular magnetic fields.
 - (B) perpendicular electric fields.
 - (C) parallel electric and magnetic fields.
 - (D) perpendicular electric and magnetic fields.
- 18** What did William and Lawrence Bragg use X-rays to investigate?
- (A) The speed of light
 - (B) The emission of photoelectrons
 - (C) The crystal structure of materials
 - (D) The charge to mass ratio of an electron

19 Why is pure copper a better electrical conductor than pure silicon?

- (A) Electrons move through copper in pairs.
- (B) Silicon contains fewer free electrons than copper.
- (C) Copper has a conduction band and silicon does not.
- (D) Copper atoms contain more electrons than silicon atoms.

20 The diagrams show possible ways to connect the coils and rotor of a DC motor to a DC power supply.

In which circuit will the rotor turn in a clockwise direction?



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Centre Number

Section I (continued)

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Part B – 55 marks

Attempt Questions 21–32

Allow about 1 hour and 40 minutes for this part

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

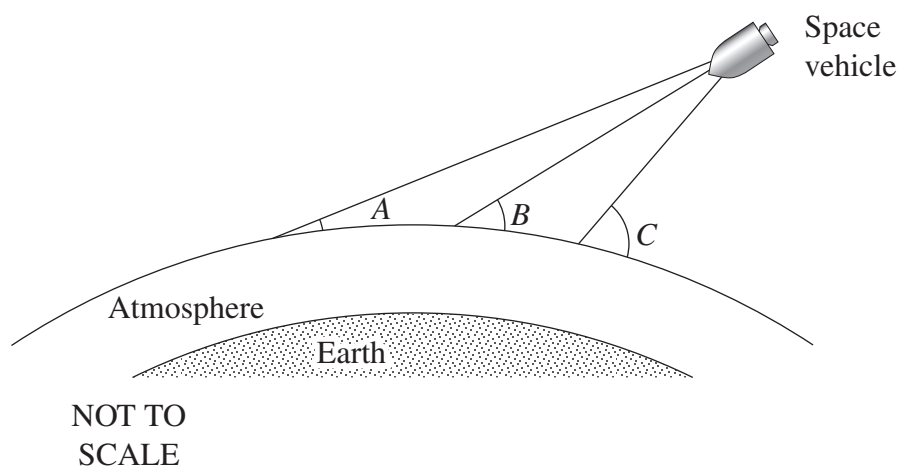
Show all relevant working in questions involving calculations.

Question 21 (2 marks)

The optimum angle for safe re-entry of a space vehicle into Earth's atmosphere is angle B .

2

Outline consequences of the space vehicle entering the atmosphere at angle A AND angle C .



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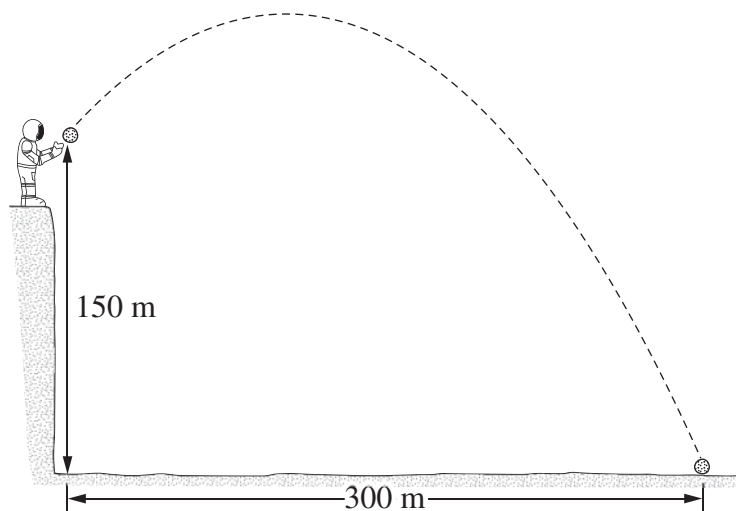
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Question 22 (5 marks)

An astronaut on the Moon throws a stone from the top of a cliff. The stone hits the ground below 21.0 seconds later. The acceleration due to gravity on the moon is 1.6 ms^{-2} .



- (a) Calculate the horizontal component of the stone's initial velocity. Show your working. **1**

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- (b) Calculate the vertical component of the stone's initial velocity. Show your working. **2**

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- (c) On the diagram, sketch the path that the stone would follow if the acceleration due to gravity was higher. The initial velocity is the same. **2**

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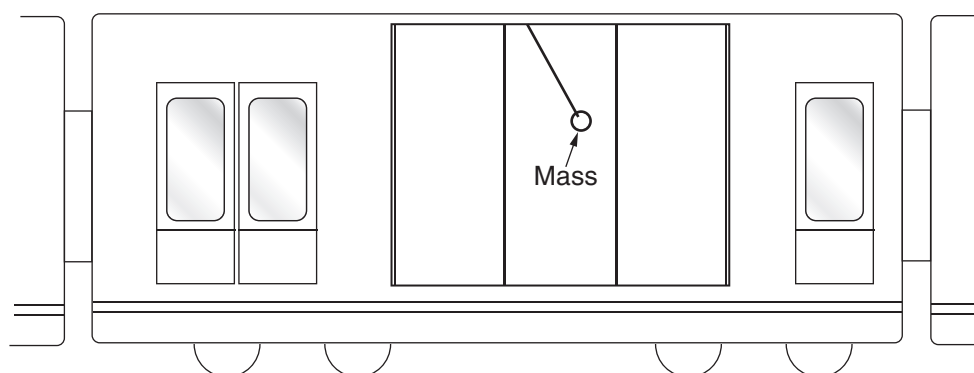
Section I – Part B (continued)

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Question 23 (5 marks)

A train is travelling on a straight horizontal track. A student on the train attaches a mass on a string to the ceiling of the train. The student observes that the mass remains stationary in the position shown.



- (a) Why does the mass hang with the string at an angle to the vertical? 2

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- (b) The string then breaks and the mass falls. 3

Indicate the path of the mass on the diagram above. Explain why the mass has taken this path.

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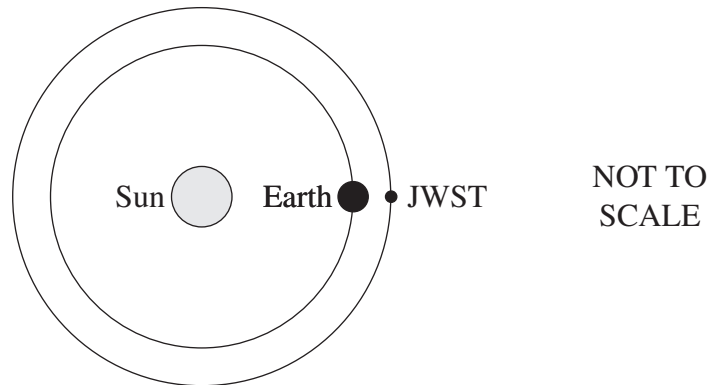
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Question 24 (3 marks)

In 2014 the James Webb Space Telescope (JWST) will be placed in orbit around the Sun. Earth and the JWST will follow the orbits shown, with identical orbital periods. This appears to contradict Kepler's law of periods.

3



Why is it possible for the JWST to orbit the Sun with the same orbital period as Earth?
In your answer, refer to Kepler's law of periods.

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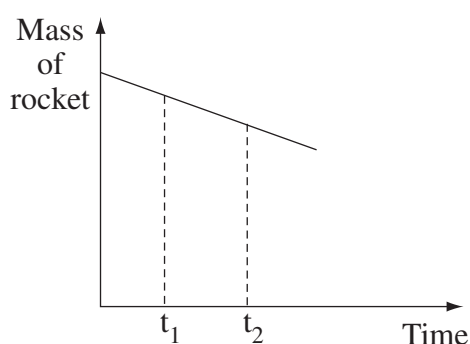
Section I – Part B (continued)

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Question 25 (4 marks)

The mass of a rocket decreases during launch as it burns fuel, as shown in the graph. The rocket engine produces a constant upward force on the rocket.



- (a) How does the law of conservation of momentum apply to the motion of the rocket? 2

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- (b) Why do the g-forces on an astronaut in the rocket differ at times t_1 and t_2 ? 2

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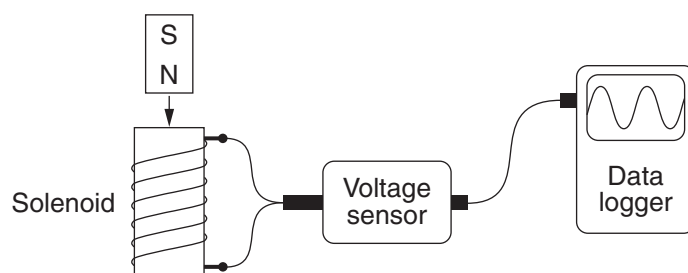
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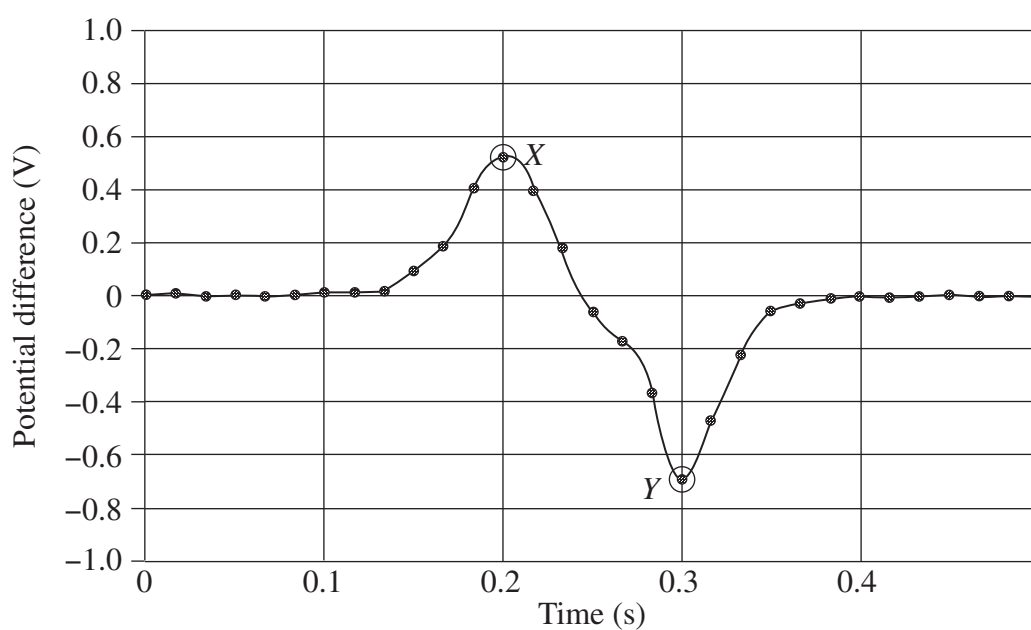
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Question 26 (5 marks)

A bar magnet is dropped through the centre of a solenoid connected to a data logger as shown.



The data are recorded in the graph as shown.



- (a) Why is the magnitude of the potential difference at *Y* greater than at *X*?

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Question 26 continues on page 19

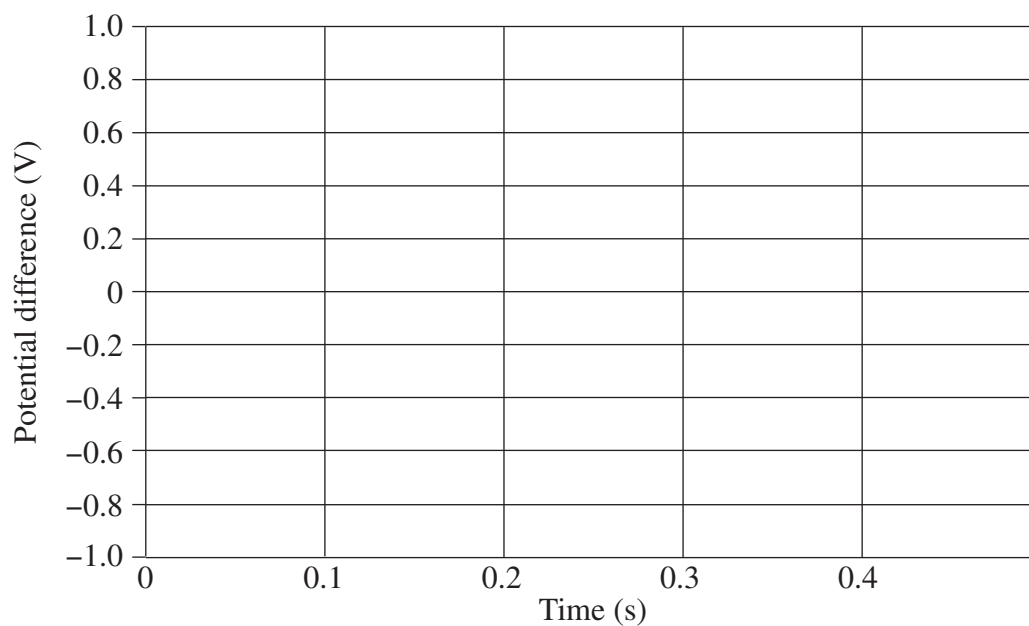
Question 26 (continued)

(b) The magnet is dropped again with two changes being made.

3

1. It is dropped from a greater height.
2. The south pole of the magnet is pointing down.

Sketch a graph that represents the most likely outcome of this new experiment.



End of Question 26

Question 27 (6 marks)

Magnetic resonance imaging is a current technology that uses superconductors. Identify two **OTHER** technologies that use superconductors. Evaluate the impact of these technologies on society and the environment.

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Centre Number

Section I – Part B (continued)

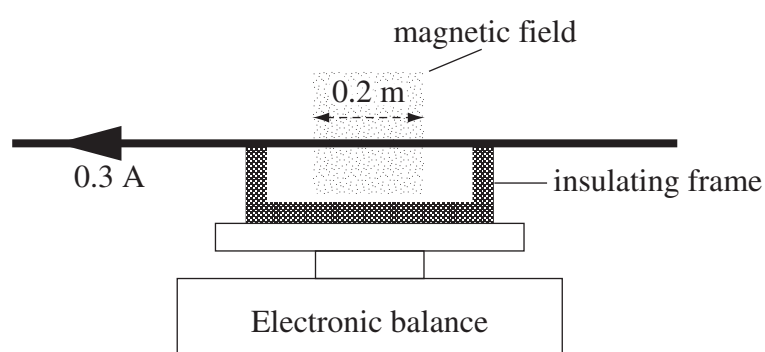
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Student Number

Question 28 (4 marks)

A copper rod is placed on a wooden frame, which is placed on an electronic balance. A length of 0.2 m of the rod passes at right angles to a horizontal magnetic field.

4



When a current of 0.3 A is passed through the rod, the reading on the balance increases by 7.5×10^{-4} kg.

What is the strength and direction of the magnetic field?

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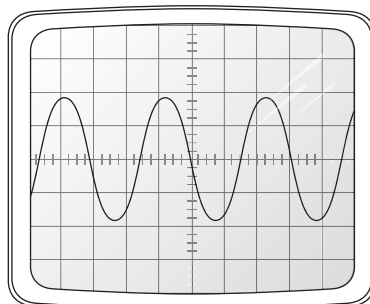
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Question 29 (3 marks)

Two sets of plates deflect an electron beam in a cathode ray oscilloscope to produce the trace on the screen as shown.

3



Explain how the deflection plates produce this pattern.

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Centre Number

Section I – Part B (continued)

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Student Number

Question 30 (5 marks)

Pure germanium can be doped by adding small amounts of boron.

- (a) Is the doped germanium an n-type or a p-type semiconductor? Justify your answer. **2**

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- (b) Explain how the addition of boron alters the electrical conductivity of germanium. **3**

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Question 31 (5 marks)

- (a) What is the energy of a photon having a wavelength of 1000 nm? **2**

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- (b) Explain why light having a wavelength longer than a certain value does not produce an electric current in a photocell. **3**

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Centre Number

Section I – Part B (continued)

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Student Number

Question 32 (8 marks)

Two significant problems that will affect a manned spaceflight to Mars are:

8

- the changes in gravitational energy
- protecting the space vehicle from high-speed electrically charged particles from the Sun.

Use your understanding of physics to analyse each of these problems.

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

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Physics

Section II

25 marks

Attempt ONE question from Questions 33–37

Allow about 45 minutes for this section

Answer the first parts of the question in a writing booklet. Answer the rest of the parts in a SEPARATE writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 33 Geophysics	28–29
Question 34 Medical Physics	30–31
Question 35 Astrophysics	32–33
Question 36 From Quanta to Quarks	34
Question 37 The Age of Silicon	35–36

Question 33 — Geophysics (25 marks)

Answer parts (a)–(b) in a writing booklet.

- (a) (i) List THREE properties of earth materials studied in geophysics. **1**
- (ii) Observations of the period of a pendulum on a ship show variations with location. **3**
- Explain how these observations could be used to deduce information about the shape of Earth. Your answer must include a description of the shape of Earth.
- (iii) The period of a pendulum is $T = 2\pi\sqrt{\frac{l}{g}}$. **2**
- The period of a particular pendulum is measured at one location as 2.00 seconds. When measured at a second location the period is 2.05 seconds.
- Determine the ratio of the strength of the gravitational field at the first location to that at the second location.
- (b) How are gravity measurements used to obtain information about the structure of Earth? **6**

Question 33 continues on page 29

Question 33 (continued)

Answer parts (c)–(d) in a SEPARATE writing booklet.

(c) Seismic waves provide information about the larger scale structure of Earth.

(i) Describe the properties of S and P waves. 2

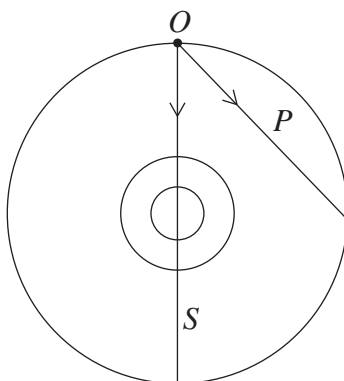
(ii) The speed of P waves is given by $v_P = \sqrt{\frac{K + \frac{4\mu}{3}}{\rho}}$. 3

The speed of S waves is given by $v_S = \sqrt{\frac{\mu}{\rho}}$.

- K , μ and ρ are all positive quantities
- ρ increases with depth
- μ , which is zero for a liquid, increases with depth at a faster rate than ρ

From this information about S and P waves, deduce how their velocities vary with depth.

(iii) In what way are the paths of the S and P waves shown below incorrect? 2



(d) There is current acceptance of the idea that the continents have moved relative to each other. 6

Why was there initial reluctance to accept this idea, and what evidence led to its acceptance?

End of Question 33

Question 34 — Medical Physics (25 marks)

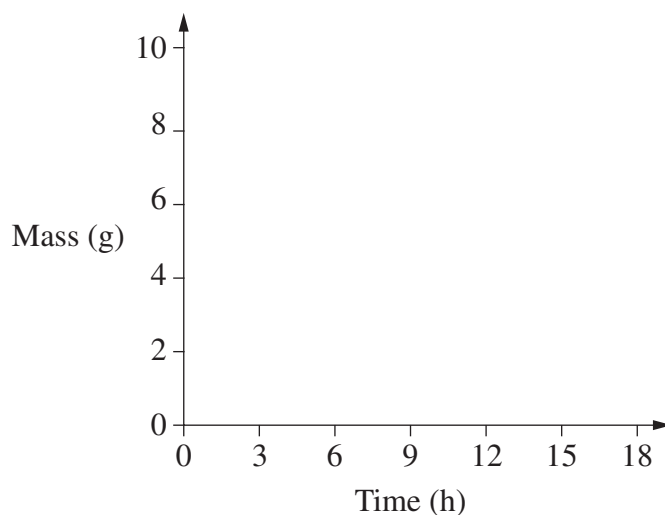
Answer parts (a)–(c) in a writing booklet.

(a) Describe how an X-ray image of the bones in the hand is produced. **3**

(b) (i) What is ONE advantage and ONE disadvantage of using a radioisotope with a 6 hour half-life for medical imaging? **2**

(ii) The half-life of Tc-99m is 6 hours. **2**

Copy the following set of axes into your writing booklet, and draw a graph to show how the mass of a 10 g sample of Tc-99m changes over 18 hours.



(c) (i) Why is hydrogen the most commonly targeted element in the magnetic resonance imaging process? **2**

(ii) Describe how energy from the radio frequency oscillator of a magnetic resonance imaging machine interacts with hydrogen to obtain information about the inside of the body. **3**

Question 34 continues on page 31

Question 34 (continued)

Answer parts (d)–(e) in a SEPARATE writing booklet.

- (d) (i) The acoustic impedance of air is $400 \text{ kg m}^{-2} \text{ s}^{-1}$. **3**

<i>Tissue</i>	<i>Density</i> (kg m^{-3})	<i>Velocity of sound</i> (m s^{-1})
Muscle	1076	1580
Bone	1912	4080
Brain	1025	1540

Calculate the intensity of the reflected ultrasound at the interface between chest muscle and air as a proportion of the incident intensity.

- (ii) Describe how the Doppler effect is used in ultrasound imaging and outline information that a Doppler ultrasound scan can provide about blood flow in the heart. **4**
- (e) A cancer specialist has access to ultrasound, CAT and PET scanners. **6**

Which of these technologies is the most appropriate to detect a 3 mm brain tumour? Justify your choice.

End of Question 34

Question 35 — Astrophysics (25 marks)

Answer parts (a)–(b) in a writing booklet.

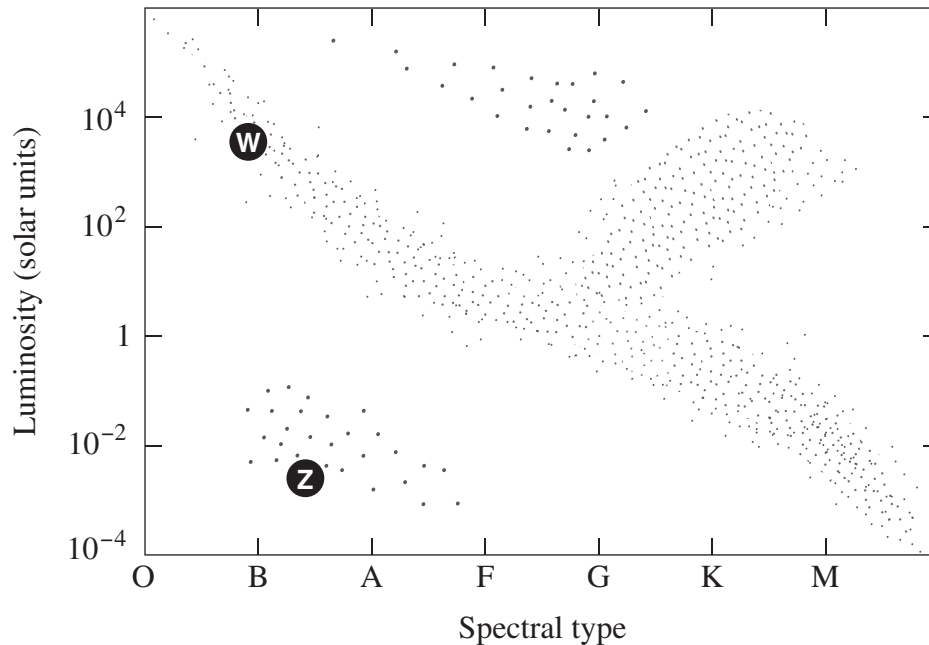
- (a) (i) The Japanese radio telescope HALCA was placed in an elliptical orbit which took it as far as 21 000 km from Earth. It took measurements simultaneously with ground-based radio telescopes. **3**
- Explain the benefit that the HALCA telescope gives radio astronomers.
- (ii) SOFIA is an infrared telescope. It is operated from high-altitude aircraft. Give TWO reasons for the greater validity of data from this telescope, compared to ground-based observations. **2**
- (b) Properties of stars, including their surface temperature and chemical composition, can be measured by using their spectra.
- (i) Identify other properties of stars which can be determined from their spectra. **2**
- (ii) Explain how surface temperatures and chemical compositions of stars can be determined from their spectra. **5**

Question 35 continues on page 33

Question 35 (continued)

Answer parts (c)–(d) in a SEPARATE writing booklet.

(c) Hertzsprung-Russell diagrams can be used to deduce the properties of stars.



- | | | |
|-------|--|----------|
| (i) | Identify the type of stars found in region Z on the above diagram. | 1 |
| (ii) | Describe the relationship between the masses of main sequence stars and their luminosities AND lifetimes. | 2 |
| (iii) | Contrast the energy production processes of stars in regions W and Z. | 2 |
| (iv) | Copy the axes from the above Hertzsprung-Russell diagram into your writing booklet. Sketch a Hertzsprung-Russell diagram for the stars in an ancient globular cluster on these axes. | 2 |
| (d) | Identify THREE advances in measurement technologies, and describe how they have improved our understanding of celestial objects. | 6 |

End of Question 35

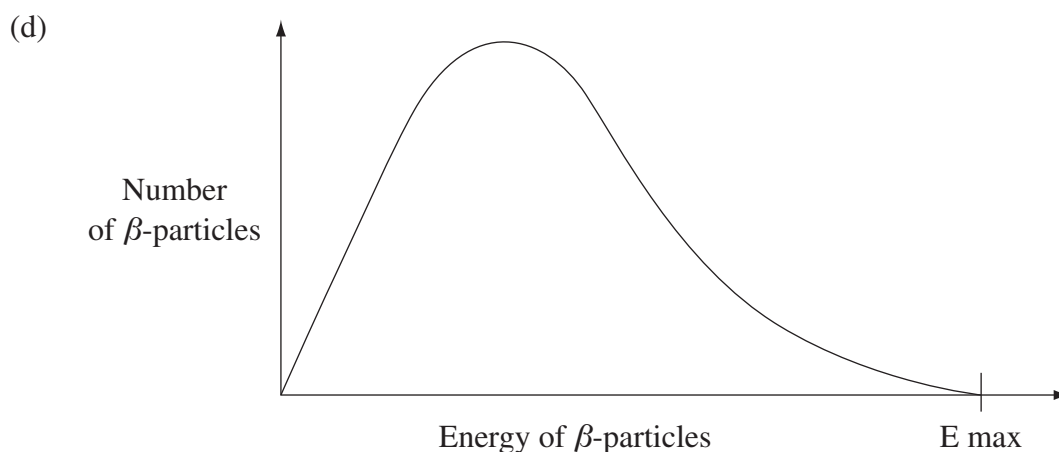
Question 36 — From Quanta to Quarks (25 marks)

Answer parts (a)–(d) in a writing booklet.

- (a) Outline the relationship between Bohr's atomic model and Planck's equation $E = hf$. **3**

- (b) Calculate the wavelength of a photon which is emitted when an electron in a hydrogen atom moves from energy level $n = 4$ to $n = 2$. **2**

- (c) How did Louis de Broglie explain the stability of electron orbits in hydrogen atoms? **3**



- How did Pauli account for the distribution of energies in the β -decay curve? **2**

Answer parts (e)–(g) in a SEPARATE writing booklet.

- (e) (i) James Chadwick discovered the neutron in 1932. How did Chadwick apply conservation laws? **3**

- (ii) Outline how the properties of neutrons make them useful in scattering experiments. **2**

- (f) Account for the existence of stable isotopes, referring to the forces which act within the atomic nucleus. **3**

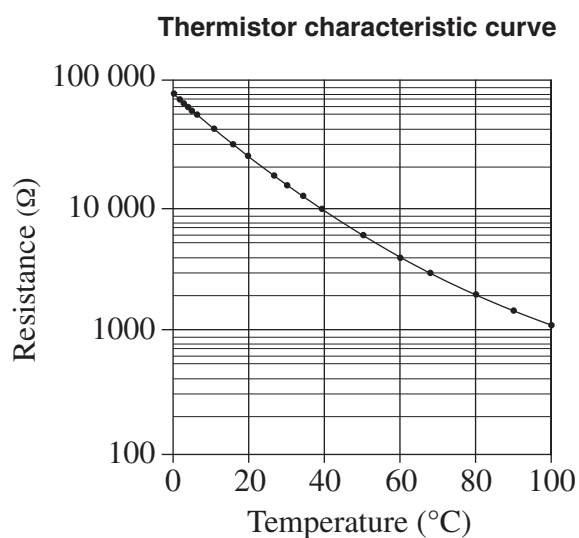
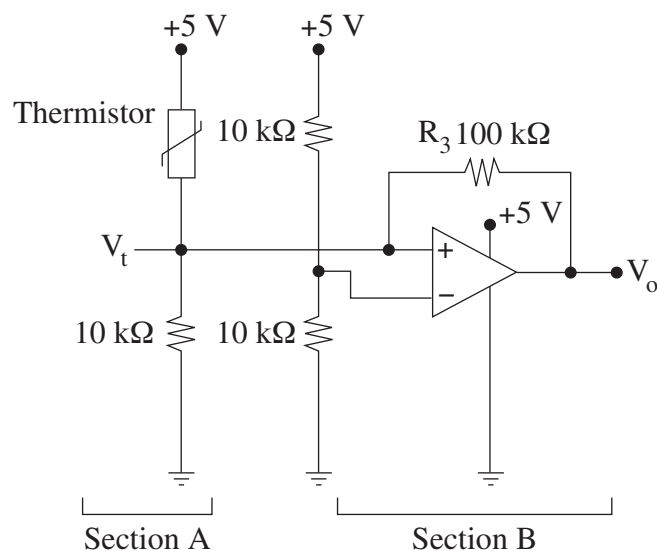
- (g) 'Important fundamental discoveries in physics often lead to applications which have a significant effect on society.' **7**

Evaluate this statement, with reference to the contributions of Rutherford, Einstein and Fermi to the development of the atomic bomb.

Question 37 — The Age of Silicon (25 marks)

Answer parts (a)–(b) in a writing booklet.

(a) A circuit and a graph of a thermistor characteristic curve are shown.

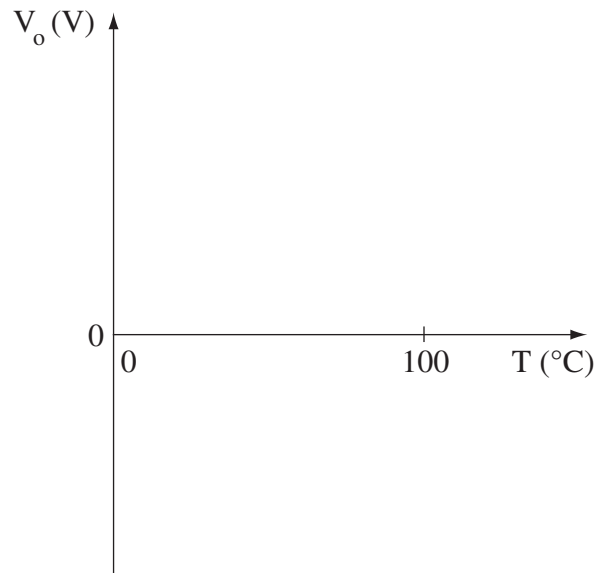


- | | |
|--|----------|
| (i) What is the function of the circuit in Section A? | 1 |
| (ii) Ignoring the rest of the circuit, what is the value of V_t at 60°C ? | 2 |
| (iii) Explain the function of the circuit in Section B. In your answer, include an explanation of the function of resistor R_3 . | 4 |

Question 37 continues on page 36

Question 37 (continued)

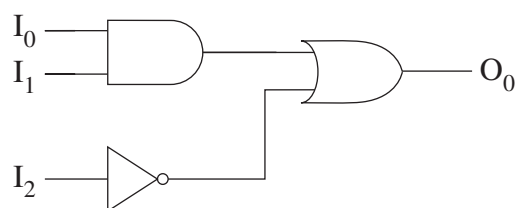
- (iv) Copy the following axes into your writing booklet, and draw a graph of the output voltage versus temperature, as temperature increases from 0°C to 100°C. 3



- (b) Outline the construction and operation of a relay. Include diagrams in your answer. 4

Answer parts (c)–(e) in a SEPARATE writing booklet.

- (c) In your writing booklet, draw a truth table for the following logic circuit. 3



- (d) (i) What is the function of an input transducer? 1
- (ii) What type of transducer is an LED? 1
- (e) Analyse the effect of the rapid development of integrated circuits on society. 6

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DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi} \right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \text{ therefore } a_{\text{av}} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = - \frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

Awaiting copyright

2010 HSC Physics Marking Guidelines

Section I, Part A

Question	Answer
1	C
2	D
3	A
4	B
5	D
6	B
7	D
8	C
9	C
10	D
11	D
12	B
13	D
14	C
15	B
16	A
17	D
18	C
19	B
20	B

Section I, Part B

Question 21

Criteria	Marks
Sketches in general terms TWO consequences of the angle not being optimal and for each, relates it explicitly to the angle	2
- States ONE consequence of the angle not being optimal and relates it explicitly to the angle OR - Identifies TWO consequences, but does not relate them explicitly to the angle	1

Question 22 (a)

Criteria	Marks
Substitutes correct values into the equation for the uniform horizontal velocity of the projectile	1

Question 22 (b)

Criteria	Marks
Substitutes the correct values, including +/- to indicate directions, into correct equation, relating initial vertical velocity to the other known variables	2
Substitutes the magnitudes of the correct variables into correct equation relating initial vertical velocity to the other known variables	1

Question 22 (c)

Criteria	Marks
Provides sketch showing path with shorter range and lower maximum height	2
Provides sketch showing path with shorter range or lower maximum height to the angle	1

Question 23 (a)

Criteria	Marks
Provides ONE or more correct explanations including direction of acceleration	2
Correctly identifies cause of mass not hanging vertically but gives incorrect direction	1

Question 23 (b)

Criteria	Marks
- Draws a path from the position of the mass on the diagram along the line of the string - Identifies the vertical component of the acceleration - Identifies the horizontal component of the acceleration	3
Provides any TWO of the above	2
Provides any ONE of the above	1

Question 24

Criteria	Marks
- States that both the Sun and the Earth exert a gravitational force on the JWST - Reasons that the resultant force is consistent with a greater orbital speed of the JWST than would be the case if the Sun's gravitational force alone were acting on the JWST - Indicates that their analysis is related to Kepler's law of periods	3
- States Kepler's law of periods - Makes a correct deduction about the motion of the JWST	2
- States Kepler's law of periods OR - Makes a correct deduction about the motion of the JWST	1

Question 25 (a)

Criteria	Marks
- States that the momentum of the burnt fuel and the momentum of the rocket must be equal but in opposite directions for total momentum to be conserved OR - Reasons that the change in momentum of the fuel and the rocket is produced by equal and opposite forces acting on the fuel and rocket	2
- States that the momentum of the burnt fuel and the momentum of the rocket are in opposite directions OR - Describes the force pair in opposite directions associated with the fuel and rocket OR - States the law of conservation of momentum	1

Question 25 (b)

Criteria	Marks
Relates the decrease in mass to an increase in acceleration	2
Relates a change in mass to acceleration	1

Question 26 (a)

Criteria	Marks
Correctly relates the rate of flux change, and hence potential difference, to the speed of the falling magnet.	2
- States that the magnet is moving faster at Y than it is at X. OR - States that the rate of flux change is greater at Y than X.	1

Question 26 (b)

Criteria	Marks
- Shows on the graph that compared to the original graph: - the induced potential difference is initially negative, and then positive - the potential difference has a greater magnitude at positions corresponding to X and Y - it takes less time for both the negative and positive changes in potential difference to take place	3
Correctly shows TWO changes to the graph that result from the described experiment	2
Correctly shows ONE change to the graph that results from the described experiment	1

Question 27

Criteria	Marks
- States TWO applications of superconductors that are not related to MRI - Provides a point for each application to show how these applications affect society AND/OR the environment - Provide a clear judgement as to the impact on society and the environment of ONE or TWO applications	5–6
- States TWO applications of superconductors that are not related to MRI - Provides a point for each application to show how the applications affect society AND/OR the environment OR - States TWO applications of superconductors that are not related to MRI - Provides a point for ONE application to show how ONE application affects society AND/OR the environment	3–4
Mentions valid applications but does not discuss their impact on society AND/OR the environment	1–2

Question 28

Criteria	Marks
States the direction of the field unambiguously and correctly calculates strength	4
- Calculates strength correctly OR - Calculates strength incorrectly by failing to convert kilograms to Newtons AND correctly states the direction of the field	3
Calculates strength incorrectly by failing to convert kilograms to Newtons	2
- States the direction of the field unambiguously OR - Provides a similar relevant formula	1

Question 29

Criteria	Marks
Correctly identifies the number of plates and how signals applied to them result in the observed pattern	3
Identifies that there are two pairs of plates which deflect the beam horizontally and vertically	2
States that the plates deflect the beam or that they produce an electric field	1

Question 30 (a)

Criteria	Marks
Identifies the semiconductor as p-type and relates this to a comparison of the number of outer electrons in boron and germanium	2
Identifies the semi conductor as a p-type and relates this to the electron structure of boron	1

Question 30 (b)

Criteria	Marks
States that the addition of boron increases the conductivity and relates this to the greater number of charge carriers resulting from hole formation, in comparison to electron-hole pair formation in pure germanium	3
Associates increased conductivity with the formation of holes	2
Identifies the conductivity of boron-doped germanium is better than pure germanium	1

Question 31 (a)

Criteria	Marks
• Selects correct formulae and correctly substitutes	2
• Correctly links relevant formulae OR • Correctly uses ONE of the formulae	1

Question 31 (b)

Criteria	Marks
• Shows a good understanding of the photoelectric effect and relates it to the production of a current • Shows understanding that longer wavelengths mean lower energy photons	3
• Shows a basic understanding of the photoelectric effect • Mentions the connection between cut-off frequency and some threshold in response OR • Shows understanding that longer wavelengths mean lower energy photons	2
• Provides a relevant statement about the photoelectric effect OR • States a connection between wavelength and photon energy	1

Question 32

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the physics related to the problems affecting a manned spaceflight to Mars • Identifies the components of each problem • Draws out or shows the relationship between the physics and the problem • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7–8
• Demonstrates sound knowledge and understanding of the physics related to the problems affecting a manned spaceflight to Mars • Describes the issues related to the problems • Communicates some scientific principles and ideas in a clear manner	5–6
• Demonstrates a basic knowledge of the physics related to either problem identified • Outlines the problems or the implications of these • Communicates ideas in a basic form using general scientific terms	3–4
• Demonstrates a limited knowledge of the physics related to either problem identified • States an aspect of the problems or an implication • Communicates simple ideas	1–2

Section II

Question 33 (a) (i)

Criteria	Marks
Identifies THREE properties studied in geophysics	1

Question 33 (a) (ii)

Criteria	Marks
- Correctly outlines the shape of Earth - Shows how the period at sea level varies with latitude - States that the period varies inversely with the value of g and that g increases if closer to the centre	3
Any TWO of the above	2
Any ONE of the above	1

Question 33 (a) (iii)

Criteria	Marks
Correct substitution into the TWO formulae to determine ratio	2
Correct substitution in the formula	1

Question 33 (b)

Criteria	Marks
Provides a coherent, extensive and detailed discussion, explaining steps in going from initial measurement to final result. Discussion may mention gravimetric surveys, steps included in data reduction, Earth properties, satellite paths, structure and function of gravimeter	5–6
Provides satisfactory but not extensive discussion of how gravity data obtained and the uses to which it is put	3–4
Shows limited understanding of how gravity data obtained and uses to which it is put	1–2

Question 33 (c) (i)

Criteria	Marks
• Provides features or characteristics of the properties of S AND P waves	2
• Provides a correct statement about ONE type of wave	1

Question 33 (c) (ii)

Criteria	Marks
• Identifies the THREE properties that can be deduced from the equations	3
• Identifies TWO of the properties	2
• Identifies ONE of the properties	1

Question 33 (c) (iii)

Criteria	Marks
• Indicates why the path of the P-wave is incorrect • Indicates why the path of the S-wave is incorrect	2
• Indicates why the path of the P-wave is incorrect OR • Indicates why the path of the S-wave is incorrect	1

Question 33 (d)

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the plate tectonics theory • Describes thoroughly the evidence supporting plate tectonics • Shows how this overcame the initial reluctance to accept the notion of continental movement • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	5-6
• Demonstrates sound knowledge and understanding of the plate tectonics theory • Describes some evidence supporting plate tectonics theory • Communicates some scientific principles and ideas in a clear manner	3-4
• Demonstrates a basic knowledge of the plate tectonic theory • Outlines a supporting piece of evidence of the plate tectonic theory • Communicates ideas in a basic form using general scientific terms	1-2

Question 34 (a)

Criteria	Marks
Relates the detection of different X-ray intensities after the X-rays pass from one side of the hand to the other with greater absorption of X-rays by bone than by soft tissues	3
Describes the passage of X-rays through the hand and either the absorption by bone or the detection by film or detector	2
Identifies ONE part of the process by which X-rays produce an image	1

Question 34 (b) (i)

Criteria	Marks
States ONE advantage AND ONE disadvantage	2
States ONE advantage OR ONE disadvantage	1

Question 34 (b) (ii)

Criteria	Marks
- Plots points accurately - Draws a smooth curved line showing a half-life of 6 hours	2
Draws a smooth curved line indicative of an exponential decay, however, points are not correct	1

Question 34 (c) (i)

Criteria	Marks
States reasons why hydrogen is the main element targeted in the MR imaging process.	2
States a reason why hydrogen is the main element targeted in the MR imaging process.	1

Question 34 (c) (ii)

Criteria	Marks
- Provides features that include: - RF energy at the Lamor frequency is directed into the body - The RF energy is absorbed by the hydrogen nuclei - RF energy is emitted from the body by the relaxing nuclei when the RF oscillator is turned off	3
- Provides features that include: - RF energy at the Lamor frequency is directed into the body - RF energy is emitted by the relaxing nuclei	2
Any correct and relevant statement	1

Question 34 (d) (i)

Criteria	Marks
- Calculates z of muscle - Substitutes acoustic impedance data into I_r/I_o formula	3
- Calculates z of muscle - Selects I_r/I_o formula	2
- Calculates z of muscle OR - Selects I_r/I_o formula	1

Question 34 (d) (ii)

Criteria	Marks
- Relates ultrasound frequency change to relative motion of source and detector - States that the frequency change is caused by movement of blood and the rate of flow can be determined from the magnitude and direction of the Doppler shift - Sketches in general terms a diagnostic use of Doppler ultrasound in the heart	4
Describes a relationship between movement and frequency change and identifies a diagnostic use of Doppler ultrasound	3
Identifies a feature of Doppler ultrasound and relevant diagnostic use	2
- Identifies a feature or use of Doppler ultrasound OR - Defines the Doppler Effect in general terms	1

Question 34 (e)

Criteria	Marks
- Identifies the most appropriate technology to use - Justifies their choice in terms of the nature of the technique and its ability to detect function - Discounts the other techniques with valid reason(s) - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	5–6
- Identifies most appropriate technology and gives a valid reason - Describes the techniques AND/OR their diagnostic uses - Communicates some scientific principles and ideas in a clear manner	3–4
- Identifies most appropriate technology and gives a reason - Outlines the techniques or their diagnostic uses - Communicates ideas in a basic form using general scientific terms	1–2

Question 35 (a) (i)

Criteria	Marks
Recognises that interferometry is being used and relates the long baseline made possible by the space telescope and ground-based radio telescopes to improved resolution	3
States that better resolution is possible using HALCA	2
States that the use of HALCA allowed a greater distance between the ground-based radio telescope and HALCA than would be possible using ground-based radio telescopes alone	1

Question 35 (a) (ii)

Criteria	Marks
Links the measurements taken at high altitude to TWO reasons for the improved validity of the data	2
States ONE reason why measurements at high altitude are better than ground-based measurements	1

Question 35 (b) (i)

Criteria	Marks
Identifies TWO properties of stars that can be deduced from their spectra	2
Identifies a property of stars that can be deduced from their spectra	1

Question 35 (b) (ii)

Criteria	Marks
Relates clearly both surface temperature and chemical composition to the corresponding features of stellar spectra	4–5
Relates both surface temperature and composition to features of the spectrum, without clearly explaining the links	2–3
Accounts for ONE of the identified properties only	1

Question 35 (c) (i)

Criteria	Marks
Identifies white dwarfs as the type of star	1

Question 35 (c) (ii)

Criteria	Marks
Provides the features of the relationship by: – Stating that as mass increases, luminosity increases – Stating that as mass increases, lifetime decreases	2
States ONE correct relationship between mass/luminosity, and mass/lifetime	1

Question 35 (c) (iii)

Criteria	Marks
Shows the difference in the energy production by: – Identifying that stars at W produce energy through fusion – Identifying that stars at Z do not produce appreciable energy through fusion OR – Identifying that stars at Z produce energy through gravitational collapse	2
- Identifies that stars at W produce energy through fusion OR - Identifies that stars at Z do not produce appreciable energy through fusion OR - Identifies that stars at Z produce energy through gravitational collapse	1

Question 35 (c) (iv)

Criteria	Marks
- Indicates that the top left region of the diagram is free of stars - Shows a linear region of stars with a clear bend to the right and up as it moves from bottom right towards the top left	2
- Indicates a region of stars with a clear bend in it OR - Indicates a region of stars in the bottom right corner of the diagram	1

Question 35 (d)

Criteria	Marks
- Identifies THREE advances in measurement technologies and demonstrates thorough knowledge and understanding of how they improve our understanding of celestial objects - Shows clearly the relationship between the advances and specific improvements in understanding that have resulted from them - Demonstrates coherence and logical progression and includes correct use of scientific principles	5–6
- Identifies TWO advances in measurement technology and demonstrates sound knowledge and understanding of advances in measurement technology and how they improve our understanding of celestial objects - Describes the measurement technology and attempts to link this to an advance in understanding OR - Identifies the technology and clearly shows how the technology improved our understanding - Communicates some scientific principles and ideas in a clear manner	3–4
- Demonstrates a basic knowledge of technology used to measure/observe celestial objects - Outlines how our understanding has improved - Communicates ideas in a basic form using general scientific terms	1–2

Question 36 (a)

Criteria	Marks
Sketches in general terms the relationship between Bohr's atomic model and Planck's equation	3
Outlines the emission or absorption of photons or the energy levels, and limited to Planck's equation	2
Provides any relevant information	1

Question 36 (b)

Criteria	Marks
• Correctly substitutes into equation	2
• Uses correct equation but transposes n_i and n_f	1

Question 36 (c)

Criteria	Marks
• Shows clearly how de Broglie explained the stability of electron orbits in hydrogen atoms	3
• Outlines aspects of de Broglie's explanation	2
• Provides any relevant information	1

Question 36 (d)

Criteria	Marks
• Identifies clearly the neutrino and the law of conservation of energy	2
• Identifies either the neutrino or the law of conservation of energy or states a 'third particle carries away some of the energy'	1

Question 36 (e) (i)

Criteria	Marks
• Clearly shows how Chadwick applied the conservation laws by outlining Chadwick's experiment	3
• Describes Chadwick's experiment and the conservation laws	2
• Outlines either Chadwick's experiment or conservation laws	1

Question 36 (e) (ii)

Criteria	Marks
• Sketches in general terms how the properties of neutrons make them useful in scattering experiments by identifying two properties	2
• Identifies ONE property	1

Question 36 (f)

Criteria	Marks
- Provides reasons for the existence of stable isotopes by: - Identifying THREE fundamental forces and their relative sizes/nature of the forces exerted OR - Identifies TWO forces, their relative sizes/nature and relates to stability of nucleus	3
- Describes TWO forces OR - Identifies all of the forces and whether they are attractive or repulsive	2
Any relevant information about the forces which operate in the atomic nucleus	1

Question 36 (g)

Criteria	Marks
- Demonstrates thorough knowledge and understanding of the contributions of Rutherford, Einstein and Fermi to the development of the atomic bomb - Provides a judgement of the statement - Describes the effect of the atomic bomb on society - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
- Demonstrates sound knowledge and understanding of the contributions of Rutherford, Einstein and Fermi to the development of the atomic bomb - Provides either a judgement of the statement or outlines the effect of the atomic bomb on society - Communicates some scientific principles and ideas in a clear manner	4–5
- Demonstrates a basic knowledge of the contribution of Rutherford AND/OR Einstein AND/OR Fermi to the development of the atomic bomb - Communicates ideas in a basic form using general scientific terms	2–3
- Demonstrates a limited knowledge of Rutherford AND/OR Einstein AND/OR Fermi - Communicates simple ideas	1

Question 37 (a) (i)

Criteria	Marks
Identifies section A's function correctly	1

Question 37 (a) (ii)

Criteria	Marks
- Determines the resistance of the thermistor at 60°C - Calculates the voltage V_t in the potential divider	2
- Determines the resistance of the thermistor at 60°C but not V_t OR - Calculates the voltage V_t in the potential divider but not using the correct resistance	1

Question 37 (a) (iii)

Criteria	Marks
- Identifies the overall function of the circuit - Shows how the circuit output changes depending on the inputs - Identifies and explains the effect of R3	4
- Identifies the overall function of the circuit - Shows how the circuit output changes depending on the inputs OR - Identifies and explains the effect of R3	2–3
Correct statement about the function of the circuit	1

Question 37 (a) (iv)

Criteria	Marks
- Identifies the output is 0 V for low temperatures - Identifies the output is 5 V for high temperatures - Determines the switching point is greater than 40°C but less than 45°C	3
- Identifies the output is 0 V for low temperatures OR - Identifies the output is 5 V for high temperatures OR - Determines the switching point is greater than 40°C but less than 45°C	1–2

Question 37 (b)

Criteria	Marks
- Identifies the components of a relay - Outlines the operation of a relay - Provides a diagram to support their answer	3–4
- Identifies the components of a relay OR - Outlines the operation of a relay	1–2

Question 37 (c)

Criteria	Marks
Determines the truth table for the circuit with correct elements	3
Draws a truth table with most elements correct	1–2

Question 37 (d) (i)

Criteria	Marks
Identifies function	1

Question 37 (d) (ii)

Criteria	Marks
Identifies type	1

Question 37 (e)

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the effect of the rapid development of integrated circuits on society - Provides an analysis that identifies that: – integrated circuits have developed rapidly (what has changed and how fast/Moore's Law) – relates the rapid development of electronics in general to ICs – shows that this development has enabled the development of new applications of electronics/ICs • Provides several examples of IC/electronics applications and clearly shows how their rapid advancement has had an impact on society • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	5–6
• Demonstrates sound knowledge and understanding of the effect of rapid development of integrated circuits on society • Describes some aspects of IC AND/OR electronics development • Describes ONE application and its impact on society • Communicates some scientific principles and ideas in a clear manner	3–4
• Demonstrates a basic knowledge of integrated circuits • Provides statements related to IC development and applications in society • Communicates ideas in a basic form using general scientific terms	1–2

Physics

2010 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I			
Part A			
1	1	9.2.2.10	H9
2	1	9.2.2.2	H6, H9
3	1	9.2.4.3.5, 9.2.4.2.4, 9.2.4.2.5, 9	H6, H12.4b
4	1	9.2.2.1, 8.4.3.2.1	H12.3c, H6, H7
5	1	9.2.2.8, 9.2.2.3.4	H6, H12.4b
6	1	9.2.4.2.3	H8, H11, H12, H14.1f
7	1	9.2.1.3.3	H6, H12.4b
8	1	9.3.2.2.5/6	H9
9	1	9.3.4.3.1	H7, H14.1f
10	1	9.3.3.2.4, 9.3.3.3.4	H7, H9
11	1	9.3.2.2.4, 9.3.2.2.5	H7, H9
12	1	9.1	H12.3c
13	1	9.4.2.2.3	H1, H10
14	1	9.4.2.2.2	H1, H8
15	1	9.4.1.2.5, 9.4.1.3.3, 9.2.2.2.8,	H9, H12.3c, H12.4b
16	1	9.4.1.2.3, 9.4.1.3.2	H9, H12.3c
17	1	9.4.1.2.8	H1, H9
18	1	9.4.4.2.1	H1, H8, H10
19	1	9.4.3.2.2, 9.4.3.2.1, 9.4.3.2.4	H9
20	1	9.3.1.2.4, 9.3.1.2.5	H9, H12.3c
Section I			
Part B			
21	2	9.2.2.2.12, 9.2.2.2.13	H7, H9
22(a)	1	9.2.2.3.1	H9, H12, H14
22(b)	2	9.2.2.3.1	H9, H12, H14
22(c)	2	9.2.2.3.1	H9, H12, H14
23(a)	2	9.2.4.3.2, 9.2.4.2.4	H6, H9, H14
23(b)	3	9.2.4.5.2	H6, H9, H14
24	3	9.2.2.2.8, 9.2.2.2.10, 9.2.2.3.5	H9
25(a)	2	9.2.2.2.7	H6
25(b)	2	9.2.2.2.7, 9.2.2.2.5	H6, H14
26(a)	2	9.3.2.3.2, 9.3.2.2.4	H9, H14
26(b)	3	9.3.2.2.4, 9.3.2.3.2	H13, H14
27	6	9.4.4.3.5	H4
28	4	9.3.1.2.1, 9.3.1.3.3	H9
29	3	9.4.1.2.9, 9.4.1.2.4	H9

Question	Marks	Content	Syllabus outcomes
30 (a)	2	9.4.3.2.7	H7
30 (b)	3	9.4.3.2.6	H7
31 (a)	2	9.4.2.2.6, 9.4.2.3.4	H7, H8, H14, H10
31 (b)	3	9.4.2.3.3	H7, H8, H10
32	8	9.2.1.2.2, 9.2.3.2.3, 9.2.2.12, 9.2.2.13, 9.3.1.2.3	H6, H7, H9

Section II

Question 33 — Geophysics

(a)(i)	1	9.5.1.2.1	H9
(ii)	3	9.5.1.3.1	H9
(iii)	2	9.5.1.3.1	H9
(b)	6	9.5.2.2.5, 9.5.2.2.6, 9.5.2.2.7, 9.5.2.2.8, 9.5.2.2.9	H9
(c) (i)	2	9.5.3.2.1	H8
(ii)	3	9.5.3.2.5, 9.5.3.2.2	H8
(iii)	2	9.5.3.3.3	H8, H14
(d)	6	9.5.4.2.4, 9.5.4.2.3	H2, H1

Section II

Question 34 — Medical Physics

(a)	3	9.6.2.2.1, 9.6.2.2.4	H8
(b)(i)	2	9.6.3.2.1	H7
(b)(ii)	2	9.6.3.2.1	H7, H13
(c)(i)	2	9.6.4.2.8, 9.6.4.2.3	H8
(ii)	3	9.6.4.2.6, 9.6.4.2.7	H7
(d)(i)	3	9.6.1.3.5	H8, H12.3c
(ii)	4	9.6.1.2.8, 9.6.1.3.3	H8
(e)	6	9.6.1, 9.6.2, 9.6.3	H4, H8, H10

Section II

Question 35 — Astrophysics

(a)(i)	2	9.7.1.2.5	H8
(ii)	2	9.7.1.2.4, 9.7.1.2.2	H14, H8
(b)(i)	2	9.7.3.2.5	H7, H10
(b)(ii)	1	9.7.3.2.3	H7, H10
(c)(i)	1	9.7.6.3.2	H14
(ii)	2	9.7.6.2.1, 9.7.6.2.2, 9.7.6.2.3, 9.7.6.3.2	H7, H9
(iii)	2	9.7.6.2.3, 9.7.6.3.2	H7, H9
(iv)	2	9.7.6.2.5, 9.7.6.3.1, 9.7.6.3.2	H13
(d)	6	9.7.4.3.3, 9.7.1.2.5, 9.7.1.2.5	H14

Section II

Question 36 — From Quanta to Quarks

(a)	3	9.8.1.2.2, 9.8.1.2.4, 9.8.1.2.5	H10
(b)	2	9.8.1.3.2	H10
(c)	3	9.8.2.2.4, 9.8.2.3.1	H8, H10

Question	Marks	Content	Syllabus outcomes
(d)	2	9.8.3.2.6	H7
(e)(i)	3	9.8.3.2.2	H6, H7, H10
(e)(ii)	2	9.8.4.2.3	H7
(f)	3	9.8.3.2.7, 9.8.3.2.8	H10
(g)	7	9.8.4.3.1, 9.8.3.2.9, 9.8.3.2.10, 9.8.3.2.11, 9.8.1.2.1	H1, H8

Section II

Question 37 — The Age of Silicon

(a)(i)	1	9.9.2.2.5	H9, H14
(ii)	2	9.9.3.2.4, 9.9.2.3.3	H9, H14
(iii)	4	9.9.6.2.3-6, 9.9.6.2.10	H9
(iv)	3	9.9.6.3.1	H9, H14
(b)	4	9.9.4.2.2	H9
(c)	3	9.9.5.3.1, 9.9.5.2.1, 9.9.5.3.2	H9, H14
(d)(i)	1	9.9.3.2.1	H9
(ii)	1	9.9.4.2.3, 9.9.3.2.1	H9
(e)	6	9.9.1.3.1, 9.9.1.2.3, 9.9.1.2.2, 9.9.7	H3, H4